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Roll No

IT-6004-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Compiler Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write down the output of each phase for the expression
position: = initial + rate*60 . 7
b) Define cross compilation and explain the process of
bootstrapping. 7
2. a) Explain the role of lexical analyzer and write in detail
about input buffering. 7
b) Consider the following grammar: 7
 $S \rightarrow 0A|1B|0|1, A \rightarrow 0S|1B|1, B \rightarrow 0A|1S$
construct leftmost derivations and parse trees for the
following sentences
i) 0101
ii) 1100101

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PTO

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3. a) Obtain the predictive parsing table for the following grammar: 7
 $S \rightarrow iEtSS' \mid a;$
 $S' \rightarrow eS \mid \epsilon;$
 $E \rightarrow b$
- b) Construct LL (1) parsing table for the following grammar:
 $S \rightarrow iCtS \mid iCtSeS \mid a;$
 $C \rightarrow b$
Is the grammar is LL (1)? 7
4. a) What is FIRST and FOLLOW? Write the algorithm for FIRST and FOLLOW. 7
- b) Consider the grammar
 $E \rightarrow TE^1$
 $E^1 \rightarrow + E/\epsilon$
 $T \rightarrow FT^1$
 $T^1 \rightarrow T/\epsilon$
 $F \rightarrow PF^1$
 $F^1 \rightarrow *F^1/\epsilon$
 $P \rightarrow (E)/a/b/\epsilon$
Compute FIRST and FOLLOW. 7
5. a) Give the syntax-directed definition for if-else statement. 7
- b) How Back patching can be used the generate code for Boolean expression and flow of control statements. 7
6. a) Explain in detail the translation of assignment statements. 7
- b) Discuss about the run time storage management. 7

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7. a) What is a three address code? Mention its types. How would you implement the three address statements? Explain with examples. 7
- b) Explain DAG representation of the basic blocks with an example. 7
8. a) Explain the principle sources of code optimization in detail. 7
- b) Define the following: 7
 - i) Code motion
 - ii) Peephole optimization
